

Endogenous Heterogeneous Gender Norms and Female Labor Supply in an Intra-Household Bargaining Model

ODD-style model description of `gender_model_shocks_preferences.nlogox`

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1 Overview

1.1 Purpose

The model explains female (and male) labor supply as the joint outcome of (a) intra-household Nash bargaining over market work and an income transfer, and (b) endogenous, heterogeneous gender norms transmitted through same-sex social networks.

Heterogeneity in wages, preferences for private vs.

public consumption, and conformism, interacted with network structure, generates dispersion and persistence in working time and transfers. Relative to the baseline (`gender_model_public_final.nlogo`), this variant adds (i) exogenous shocks at `shock-start` — a wage cut for directly affected households and/or a downward shift of preference-private — with gradual exponential recovery (`shock-depreciation`); (ii) continuous endogenous preference adaptation toward observed working time (`update-preferences`, weight `lambda`); (iii) an (inactive-by-default) gender-gap-closing wage-growth process (`update-wages`); and (iv) explicit shock-propagation observers (directly affected, their network neighbours, and household employment types). Comparative experiments vary the shock type, size, reach (`perc-affected-*`), recovery speed, preference baselines, conformism, utility specification, and network topology (random, Watts–Strogatz, preferential attachment, similarity-based, homophily, none, homogenous mixing).

The implementation analysed here is `gender_model_shocks_preferences.nlogox`, procedures `setup`, `go`, `start-shock`, `end-shock`, `set-theta`, `choose-bundle`, `calculate-utility`, `calculate-payoff`, `update-preferences`, `update-wages`, `update-statistics`.

1.2 Entities, state variables, and scales

There are two agent breeds plus two link/collection entities:

Entity	State variables	Notes
women men (turtle breeds)	current-working-time in [0,1] old-working-time current-transfer-to-woman theta in [-1,1] old-transfer-to-woman spouse (partner pointer) wage >= 0 wage-pre (pre-shock wage memory)	One woman + one man = one household. number only used in import-csv mode. memory-change is solver scratch state. wage-pre/preference-private-pre are shock baselines for recovery. norm-parameter/perception-norm-division-of-labor were locals in the

Entity	State variables	Notes
	<pre> preference-private in [0,1] preference-private-pre (pre-shock preference memory) preference-difference = h - alpha (diagnostic) conformism >= 0 current-utility, memory-change, number norm-parameter, perception-norm- division-of-labor (stored norm penalty/percept) </pre>	baseline and are now turtles-own so they can be traced/exported.
Household (implicit)	(woman, man, theta) triple; theta>0 = transfer man→woman	No separate NetLogo agent; bargaining unit in set-theta. Outside option defined at theta=0.
Social links (links breed)	Undirected same-sex ties	Single breed shared by both sexes but only linked within sex, except via spouse-of-neighbour perception.
Observer (globals)	<pre> global-mean-working-time-men/ women, global-mean-transfer, history-*, average-mean-*, sub- group agentsets run-time, current_wage_gap directly-affected-men/women, neighbors-of-affected-men/women + mean-current-working-time-* single-earner-*, dual-earner-*, both-unemployed-* + mean-current- working-time-* </pre>	Median-split preference/wage subgroups fixed at setup as before. Shock-propagation and household-type agentsets also fixed at setup from the initial directly-affected-* draw; histories are 10-period moving windows.

Scales. No explicit space matters: layout-circle is visualisation only. Time is discrete ticks; one go = shock check + household re-bargaining + norm updating + preference adaptation. shock-start defaults to 60; recovery runs for all later ticks. BehaviorSpace runs use 1000–4000 ticks (time limits 1000–4000). Labour and transfer are normalised fractions, not hours.

1.3 Process overview and scheduling

Phase	What happens
setup	Clear, reset-timer, seed RNG if fixed-rs. Generate same-sex networks (for random-network, watts-strogatz, preferential-attachment). Create agents, pair spouses, draw wage / preference-private / conformism / theta from truncated normals around slider means. Optionally override one specific-couple or overwrite from CSV. Fix median-split subgroup agentsets. Optionally build similarity networks (preference-private, wage, conformism, homophily). Initialise global-mean-working-time-*. Draw directly-affected-women/men via n-of (N * perc-affected-*/100); derive neighbors-of-affected-* from link-neighbors and household-type sets (single-earner-*, dual-earner-*, both-unemployed-*). reset-ticks.
go, shock block	If ticks = shock-start, run start-shock once; if ticks > shock-start, run end-shock every tick (partial recovery toward wage-pre / preference-private-pre at rate shock-depreciation).
go, set-theta	For each woman in sequence: (1) compute outside options via choose-bundle(self, spouse, 0); (2) hill-climb theta up then down in steps of 0.001

Phase	What happens
	using calculate-payoff; (3) commit best-theta and the associated choose-bundle labour pair to both spouses.
go, choose-bundle	Iterated best response: woman and man alternately hill-climb own h by current-delta = 0.0001 while partner h is held fixed, until joint change <= convergence_epsilon (default 0.001). Evaluates calculate-utility.
go, update-preferences	Every tick, including shock = "no" runs: preference-difference = h - alpha; then alpha <- (1-lambda) * alpha + lambda * h, clipped to [0,1]. With default lambda = 0.002 this is slow endogenous preference formation toward own behaviour.
go, update-statistics	Copy current-* to old-*. Recompute global means, 10-period histories and moving averages. Recompute means for directly-affected-*, neighbors-of-affected-*, and household-type sets.
go, tick + I/O	Advance clock; optionally append per-turtle row via print-agent-positions; set run-time = timer. update-wages exists but is commented out in go.

Ordering is sequential over households within a tick. Norm perception uses lagged old-* values, so intra-tick ordering does not directly enter norms, but choose-bundle reads partner current-working-time while it is being mutated.

2 Design concepts

Concept	Realisation in this model
Basic principle	Collective household model with Nash bargaining + conformity-augmented CES/additive utility + social norms, extended with exogenous shocks and adaptive preferences. Norms are local means, hence heterogeneous and endogenous; preferences are now also endogenous via lambda.
Emergence	Gender gaps in hours, transfer distributions, norm clusters, and shock-propagation gradients (affected vs. neighbours vs. rest) emerge from bargaining + network topology + adaptation; not imposed beyond initial means norm-paid-time-male/female and the shock-start intervention.
Adaptation	Households adapt h_w, h_m, theta by myopic hill-climbing. Preferences adapt toward own hours via lambda. Wages/shock memories recover toward pre-shock baselines via shock-depreciation. No forward-looking search, no link rewiring after setup.
Objectives	Maximise own calculate-utility (labour stage) and Nash product ($U_m - U_{out_m}$) ($U_w - U_{out_w}$) (transfer stage). Individual rationality required, else payoff -1.
Learning / Prediction	None, except backward-looking preference assimilation (alpha follows h) and lagged imitation via norm penalty. No expectation formation.
Sensing	Agents sense: own + spouse wage/preference/conformism; partner current h; mean old-h and old-theta of same-sex link-neighbors; mean old-h of spouses-of-neighbors. If isolated: own lag, or global mean under homogenous mixing. Stored in perception-norm-division-of-labor / norm-parameter for inspection.
Interaction	Direct: intra-household bargaining. Indirect: inter-household via norm penalty $\exp(\text{norm-parameter})$. Shocks propagate indirectly through norms and bargaining; no market clearing.

Concept	Realisation in this model
Stochasticity	Initial draws (truncated normals), spouse matching one-of, network generators, rnd:weighted-n-of noise ($N(0,0.001)$), and n-of sampling of directly-affected-*. Reproducible via fixed-rs + random-seed-fixed.
Collectives	Household (bargaining unit); same-sex neighbourhood (reference group); median-split subgroups for observation only; shock-propagation sets (directly-affected-*, neighbors-of-affected-*) and employment-type sets (single-earner-*, dual-earner-*, both-unemployed-*) for observation only.
Heterogeneity	Continuous heterogeneity in wage, preference-private, conformism; structural heterogeneity via degree/position; treatment heterogeneity via affected/neighbour status.
Observation	Means/medians/SD/p25/p75 of utility, hours, transfers; subgroup means by wage/preference; moving averages; shock-propagation means and current_wage_gap; preference-difference time series/histograms; per-agent CSV trace (now with spouse hours, norm-parameter, perceived norm); time-series plots + histograms.

3 Details

3.1 Initialization (setup, set-initials-*)

1. ca; reset-timer; set current-delta = 0.0001 (was 0.00001 in baseline), clear histories.
2. If fixed-rs, random-seed random-seed-fixed.
3. Build random-network / watts-strogatz / preferential-attachment networks via nw extension **before** agents get traits; other modes create unlinked turtles first.
4. layout-circle (visual only); re-seed if fixed-rs so trait distributions are comparable across topologies.
5. Pair each woman with one-of men with [spouse = 0]; reciprocal spouse pointers.
6. Women: h = norm-paid-time-female/100, theta ~ $N(\text{initial-transfer}, \text{initial-transfer} * \text{std-dev-values})$ clipped to [0,1], wage ~ $N(\text{wage-female}, \text{mean}(\text{wage-male}, \text{wage-female}) * \text{std-dev-values})$ clipped at 0, conformism ~ $N(\text{conformism-female}, \dots)$ clipped at 0, preference-private ~ $N(\text{preference-private-mean-female}, \dots)$ clipped to [0.01,0.99]. Men symmetric with male sliders and norm-paid-time-male/100; man inherits initial theta from spouse.
7. Optional specific-couple override; median splits for high/low public-goods households and above/below-median wages (static snapshots).
8. Zero mean-current-working-time-directly-affected-* / neighbors-of-affected-*; optional similarity/homophily network construction.
9. Initialise global-mean-working-time-men/women from initial hours (used as perceived norm under homogenous mixing).
10. Draw directly-affected-women = n-of ($N * \text{perc-affected-female}/100$) women (default 30%) and directly-affected-men (default 0%); build neighbors-of-affected-* as non-affected link-neighbors of affected turtles; build household-type sets by cross-tabulating own vs. spouse affected status.
11. Optional CSV overwrite (initialTransfer.csv, moderatePreferences.csv, highConformity.csv, moderateConformity.csv — note: preferences file is read twice, second read overwrites the first).
12. reset-ticks; optionally open agent_position_<exp>_<run>.csv with extended header agentid,agenttype,period,work

Key defaults in the Interface tab: wage-female=0.9, wage-male=1.0, norm-paid-time-male=77, norm-paid-time-female=36, number-agents-each-type=400 (experiments use 10–500), preference-private-mean-male=0.45, preference-private-mean-female=0.48, initial-transfer=0.0, std-dev-values=0.2, conformism-male=10.0, conformism-female=10.0, utility-function=CES with CES_beta=0.5, weight-working-time-self/partner/transfer=1, watts-strogatz-neighbors=1, preferential-attachment-min-degree=1, convergence_epsilon=0.001, lambda=0.002, shock=no (chooser default index 3), shock-start=60, shock-depreciation=0.005, preference-private-shock-male/female=0.0, perc-affected-female=30, perc-affected-male=0, wage-growth-rate=0.0.

3.2 Input data

No time-series input. `import-csv=true` replaces random traits with files listed above. CSV handling assumes row order matches household number and provides no existence/length checks. All BehaviorSpace experiments shipped with the file use `import-csv=false`.

3.3 Submodels

3.3.1 Norm perception (calculate-utility)

Let h_i be own hours, h_j spouse hours, θ transfer-to-woman.

$$N_i^h = \text{mean old-h of link-neighbors of } i$$

$$N_i^\theta = \text{mean old-theta of link-neighbors of } i$$

$$N_i^{h_j} = \text{mean old-h of spouses-of-neighbors}$$

$$\text{norm-penalty}_i = -c_i \cdot \left(\omega_s (h_i - N_i^h)^2 + \omega_\theta (\theta - N_i^\theta)^2 + \omega_p (h_j - N_i^{h_j})^2 \right)$$

c_i is own conformism; ω are the three weight-* sliders. Isolated agents use own lag; under homogenous mixing they use the global means instead. Unlike the baseline, the percept perception-norm-division-of-labor and the penalty norm-parameter are stored back to the calling turtle (`set`, not `let`), so they persist for plots, histograms, and the per-agent CSV. Note the stored values belong to whichever agent last called `calculate-utility` on that turtle (including counterfactual `choose-bundle/calculate-payoff` evaluations), not necessarily the last committed bundle.

3.3.2 Material utility and conformity multiplier

Define private consumption x_i and domestic good $Q = 2 - h_i - h_j$:

- If recipient (relevant-transfer < θ , sign-flipped for women): $x_i = h_i w_i + |\theta| w_j h_j$.
- If donor: $x_i = h_i w_i (1 - \theta)$.
- If $x_i \leq 0$ or $h > 1$: utility = -200 (infeasibility penalty).

Four utility-function options (with α_i = preference-private, β = CES_beta):

- additive: $(\alpha_i \sqrt{x_i} + (1 - \alpha_i) \sqrt{Q}) \cdot \exp(\text{norm-penalty}_i)$.
- CES: $(\alpha_i x_i^\beta + (1 - \alpha_i) Q^\beta)^{\frac{1}{\beta}} \cdot \exp(\text{norm-penalty}_i)$.
- multiplicative: $\sqrt{x_i} \sqrt{Q} \cdot \exp(\text{norm-penalty}_i)$.
- multiplicative including weights: intended $x_i^{\alpha_i} Q^{1-\alpha_i} \cdot \exp(\dots)$; recipient branch matches this intent, donor branch as shipped computes $((x_i^{\alpha_i} \cdot Q))^{1-\alpha_i}$ — see code critique.

3.3.3 Labour best response (choose-bundle)

Holding θ fixed, each partner steps h_i by $\pm$ current-delta (0.0001) while it strictly improves own utility and stays in $[0, 1]$. Partners alternate until joint absolute change `change-a + change-b` $\leq$ `convergence_epsilon` (default 0.001). `memory-change` (last 10 steps) and `sum-memory-change` guard against 2-cycles. No step-size annealing is active in the shipped version (the `current-delta / 2` block is commented out). Relative to the baseline (> 0 exact convergence, step 0.00001), this converges earlier and more coarsely.

Julia-port note. The NetLogo description above stays factual for the reference implementation. The Julia port (GenderNorms) replaces the discrete `current-delta` hill-climb by continuous best responses in two accuracy tiers: for eligible utility regimes (additive, multiplicative, multiplicative-with-weights, and CES with $0 < \beta \leq 1$ under the stated parameter guards) each best response first checks a two-probe sign certificate at the `current-delta` resolution (0.0001) and keeps the current hours when it encloses the optimum there (the tier-1 seed certificate, one probe at a boundary step); otherwise it solves the first-order condition $d \log \frac{U}{d} h_i = 0$ over $[0, 1]$ with a safeguarded derivative root solve (bracketed Newton/bisection, argument tolerance 10^{-8}). Every ineligible or numerically failed case falls back bitwise to a seeded continuous Brent search whose argument resolution is `current-delta` and whose window half-width is 0.05 . The alternating loop, `convergence_epsilon`, and the `keep-current-hours` rule on no feasible sample are unchanged (see MDR-0017, which supersedes MDR-0002).

3.3.4 Transfer bargaining (set-theta, calculate-payoff)

For a household (w, m) let the predetermined state in tick t be wages (w_w, w_m) , preferences (α_w, α_m) , conformism (c_w, c_m) , and lagged norms N (same-sex means of old-h / old-theta plus spouses-of-neighbours means). For fixed θ , choose-bundle returns the labour subgame outcome as the fixed point of alternating best responses:

$$(h_w^*(\theta), h_m^*(\theta)) = \text{choose-bundle}(w, m, \theta), \quad h_i^*(\theta) \in [0, 1],$$

where each h_i^* maximises own calculate-utility $U_i(h_i, h_j, \theta; w_i, w_j, \alpha_i, c_i, N)$ holding partner hours and θ fixed (hill-climb by current-delta, see above). Indirect utilities conditional on θ are then:

$$\begin{aligned} U_w(\theta) &= U_w(h_w^*(\theta), h_m^*(\theta), \theta), \\ U_m(\theta) &= U_m(h_w^*(\theta), h_m^*(\theta), \theta). \end{aligned}$$

θ is the transfer-to-woman in $[-1, 1]$: $\theta > 0$ is a transfer man→woman (the woman is recipient, relevant-transfer = -theta < 0); $\theta < 0$ reverses the flow. It enters both material consumption $x_i(\theta)$ and the norm penalty on $(\theta - N_i^\theta)^2$.

Outside options are the indirect utilities at zero transfer, evaluated through the same labour response:

$$\begin{aligned} U_w^0 &= U_w(h_w^*(0), h_m^*(0), 0), \\ U_m^0 &= U_m(h_w^*(0), h_m^*(0), 0). \end{aligned}$$

With $\Delta_i(\theta) = \text{precision}(U_i(\theta), 8) - \text{precision}(U_i^0, 8)$, the Nash product implemented in calculate-payoff is:

$$\text{payoff}(\theta) = \begin{cases} (\Delta_w(\theta) \cdot \Delta_m(\theta)) & \text{if } \Delta_w(\theta) \geq 0 \text{ and } \Delta_m(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

so the household solves, in exact terms,

$$\max_{\theta \in [-1, 1]} (U_w(\theta) - U_w^0)(U_m(\theta) - U_m^0) \quad \text{s.t.} \quad U_w(\theta) \geq U_w^0, U_m(\theta) \geq U_m^0,$$

with value -1 assigned to individually irrational θ and with the 8-decimal rounding above creating flat plateaus. Labour (h_w^*, h_m^*) adjusts endogenously at every evaluated θ , i.e. this is bargaining over θ anticipating the labour Nash equilibrium, not over the triple (h_w, h_m, θ) jointly.

Numerically, set-theta approximates this with two one-sided hill-climbs from the status quo θ_t with step $\delta_\theta = 0.001$ over $[-1, 1]$, stopping at the first decrease per side:

The θ with the highest payoff and its choose-bundle hours are committed. Note: best-payoff is initialised to 0 and best-theta to 0, so the status quo is never evaluated and a no-improvement search defaults to theta=0. (Unchanged from baseline; set-theta-index exhaustive-grid alternative is still present but unused.)

3.3.5 Shocks (start-shock, end-shock)

At ticks = shock-start, start-shock fires once; for all later ticks end-shock fires:

- shock = "preference" or "both": save preference-private-pre = preference-private for all turtles, then preference-private <- max(0, preference-private - preference-private-shock-*) by sex (preference-private-shock-male/female, Interface defaults 0.0). Note this hits **all** men/women, not only directly-affected-.*.
- shock = "wage" or "both": save wage-pre = wage for all turtles, then wage <- 0.6 * wage-pre (40% cut) for turtle-set directly-affected-men directly-affected-women only.
- shock = "no": neither fires meaningfully.
- Recovery (end-shock): wage <- wage + shock-depreciation * (wage-pre - wage) for wages below baseline; preference-private <- preference-private + shock-depreciation * (preference-private-pre - preference-private) for preferences below baseline. With default shock-depreciation = 0.005 this is slow exponential decay of the shock gap. There is no overshoot: once the gap closes, updating stops.

update-preferences runs concurrently (see below), so post-shock preferences are pulled both toward preference-private-pre (recovery) and toward current h (adaptation).

3.3.6 Endogenous preference adaptation (update-preferences) and wage growth (update-wages)

- update-preferences (called every go): preference-difference = current-working-time - preference-private; preference-private <- (1-lambda) * preference-private + lambda * current-working-time, clipped to [0,1]. Default lambda = 0.002. This operates even when shock = "no", making preferences endogenous in all runs.
- update-wages (defined but commented out in go): current_wage_gap = 1 - mean(wage_women)/mean(wage_men); wage_women <- wage_women * (1 + daily-growth) with daily-growth = (1 + wage-growth-rate)^(1/365) - 1 while the gap is positive. Interface default wage-growth-rate = 0.0. An older commented-out wage rule remains inside update-statistics.

3.3.7 Network formation (generate-network, generate-homophilic-network)

Similarity networks connect each turtle to preferential-attachment-min-degree (default 1) others sampled by rnd:weighted-n-of with weight $1/|x_{\text{myself}} - x_{\text{other}} - N(0, 0.001)|$ for $x \in \{\text{preference-private, wage, conformism}\}$. Homophily uses $1/(|\delta \text{ conformism}| + |\delta \text{ wage}| + |\delta \text{ preference}|)$. Ties are created once at setup and never updated.

3.3.8 Statistics (update-statistics)

Copies current-* to old-*, recomputes global means, pushes them onto 10-element histories, and reports their moving averages average-mean-working-time-{women,men}. global-mean-transfer is the contemporaneous mean over women. Additionally, if the respective agentsets are non-empty, updates mean-current-working-time-directly-affected-{men,women}, mean-current-working-time-neighbors-of-affected-{men,women}, and the eight household-type means (single-earner-women-women/men, single-earner-men-women/men, dual-earner-women/men, both-unemployed-women/men; 0 if empty). Also sets run-time = timer in go after tick.

4 Parameters, experiments, and output

Parameter	Default	Meaning
number-agents-each-type	400	Households per sex; experiments use 10–500.
wage-{female,male}	0.9 / 1.0	Mean wage offers; dispersion via std-dev-values.
preference-private-mean-{male,female}	0.45 / 0.48	Mean weight on private vs. domestic good.
conformism-{female,male}	10.0 / 10.0	Mean norm sensitivity (was 0.0 / 3.0).
std-dev-values	0.2	Relative SD scaler for all traits.
norm-paid-time-{female,male}	36 / 77	Initial hours in percent (was 20 / 80).
initial-transfer	0.0	Mean initial θ (was 0.1).
utility-function, CES_beta	CES, 0.5	Aggregator and substitution parameter.
weight-working-time-self, weight-transfer, weight-working-time-partner	1 / 1 / 1	$\omega_s, \omega_\theta, \omega_p$.
network-structure	watts-strogatz	Same 9 options as baseline. Interface cursor on Watts–Strogatz; experiments fix it.
random-network-prob, watts-strogatz-neighbors/rewiring,	0.01, 1 / 0.1, 1	Topology hyperparameters (neighbours/degree were 2 in baseline).

Parameter	Default	Meaning
preferential-attachment-min-degree		
convergence_epsilon	0.001	New: joint-change stopping tolerance in choose-bundle (baseline: exact > 0).
lambda	0.002	New: preference-adaptation weight toward own hours.
shock	no	New chooser: wage, preference, both, no. Preference shock is global; wage shock hits directly-affected-* only.
shock-start, shock-depreciation	60, 0.005	New: tick of one-off shock; per-tick exponential recovery rate.
preference-private-shock-{male,female}	0.0 / 0.0	New: downward shift of preference-private by sex.
perc-affected-{male,female}	0 / 30	New: % of each sex in directly-affected-* (wage-shock target + observer).
wage-growth-rate	0.0	New: annual growth closing the gender wage gap in update-wages (currently inactive).
fixed-rs, random-seed-fixed	off, 10	Reproducibility switch and seed (was on/1).
specific-couple, *-in-question	off	Single-household probe overrides.
import-csv, output-locations	off	CSV trait import; per-agent trace export (extended columns).

Shipped BehaviorSpace: 50 experiments, all runMetricsEveryStep=true (baseline was false), time limits 1000–4000 ticks: experiment_shock_conformism_1–10, experiment_shock_ofat_1–10, experiment_shock_policy_1–10, experiment_all_parameters_1–10 (one seed each, Watts–Strogatz, CES, N=100–400, sweeping shock, shock-depreciation, preference baselines, conformism up to 500, CES_beta), plus test_n, experiment_test, experiment_rho/conformism/CES/strength*/epsilon/equilibrium diagnostics. Core metrics (42 in shock experiments): mean/median/SD/p25/p75 of utility, hours, transfers, subgroup means by wage/preference, mean-current-working-time-directly-affected-*, neighbors-of-affected-*, household-type means, and current_wage_gap. experiment_all_parameters_* / test_* variants track only 1–3 metrics and sweep rho (unbound slider reference), CES_beta, convergence_epsilon, and population size.

5 Known implementation quirks (for replicators)

1. Specific-woman conformism override uses the male probe value (unchanged).
2. Preferences CSV block is duplicated (unchanged).
3. set-theta never evaluates staying put; defaults to 0 on failure (unchanged).
4. multiplicative including weights donor branch parenthesisatation differs from the Cobb–Douglas intent (recipient branch is correct); unchanged from baseline.
5. precision × 8 in calculate-payoff creates plateaus at the 1e-5 labour step (unchanged; labour step is now 1e-4 with convergence_epsilon = 0.001 early stopping).
6. calc-pct indexing is only safe for interior percentiles (used as 25/75; unchanged).
7. Subgroup agentsets and Info-tab documentation are static/stub as shipped (unchanged).
8. both-unemployed-men duplicates the dual-earner-men definition (not affected + spouse not affected) instead of member? ... and member? ...; both-unemployed-women is correct.
9. Asymmetric shock reach: wage shock targets directly-affected-* only, preference shock shifts **all** turtles of each sex regardless of affected status.
10. update-preferences runs every tick even with shock = "no", so lambda > 0 always makes preferences endogenous; shock recovery and preference adaptation pull in different directions after a preference shock.

11. `perception-norm-division-of-labor / norm-parameter` persistence: stored values reflect the last `calculate-utility` call (including counterfactual evaluations), not necessarily the committed bundle.
12. `experiment_all_parameters_*`, `test_*`, and `experiment_rho/conformism/...` reference a ρ slider that does not exist in the Interface tab; they will not run as shipped without adding it (presumably the intended λ).

ODD protocol after Grimm et al. (2006, 2010, 2020). This description was reverse-engineered from code; ambiguous points are noted above rather than resolved. Source: `gender_model_shocks_preferences.nlogo` + Interface/BehaviorSpace sections.